

Segmented execution: one segment executing at a time

GPT decoder — four segmentation axes

E — embedding
split d_{model} , concat

A — attention
head groups; W_o its own segment

M — MLP
 d_{ff} chunks, running-sum

H — output head
vocab slices, streamed cross-entropy



STREAMED modes — constrained device

1 active segment
the only weights on device

+ tiny shared norms / biases
peak $\approx$ one segment (memory floor)

RESIDENT modes — constrained device

executing

all segment weights stay on device;
activations of one segment at a time

evict



load



park



backing store
GPU: host RAM | CPU: disk

streamed: all other segments,
gradients, Adam state, records
resident: gradients + Adam state

dials: recomputation · weight streaming · update schedule · dropout · KV cache $\Rightarrow$ 12 training + 4 inference modes

every mode computes the same model — identical weights, only the memory schedule changes